High school · sequences

Telescoping sums

Worked solutions

Exercise 1

Find $\sum[\frac{1}{k}-\frac{1}{k+1}]$ from $k=3$ to 10.

Interior terms cancel: $\frac{1}{3}-\frac{1}{11}=\frac{8}{33}$.

Exercise 2

Express $\frac{1}{k(k+2)}$ as a difference of two fractions.

$\frac{1}{k(k+2)}=(\frac{1}{2})[\frac{1}{k}-\frac{1}{k+2}]$.

Exercise 3

Find $\sum(2k+1)$ from $k=0$ to n using $(k+1)^2-k^2$.

$2k+1=(k+1)^2-k^2$. The sum is $(n+1)^2-0^2=(n+1)^2$.

Exercise 4

If $S_n=\frac{n}{n+1}$, find the least integer $n\geq 1$ with $1-S_n<0.01$.

$\frac{1}{n+1}<\frac{1}{100}$ means $n+1>100$, hence $n\geq 100$. The minimum is 100.

Problem: complete reasoning

Let $T_n=\sum\frac{1}{k(k+2)}$ for k from 1 to n. Derive an exact formula and find its limit. Find the least n such that the gap to the limit is strictly less than 0.01.

After decomposition, all except the first two and last two terms cancel: $T_n=\frac{3}{4}-\frac{1}{2(n+1)}-\frac{1}{2(n+2)}$. The limit is $\frac{3}{4}$. The gap decreases with n. At $n=98$ it is $\frac{1}{198}+\frac{1}{200}>0.01$; at $n=99$ it is $\frac{1}{200}+\frac{1}{202}<0.01$. Therefore the minimum is 99.